

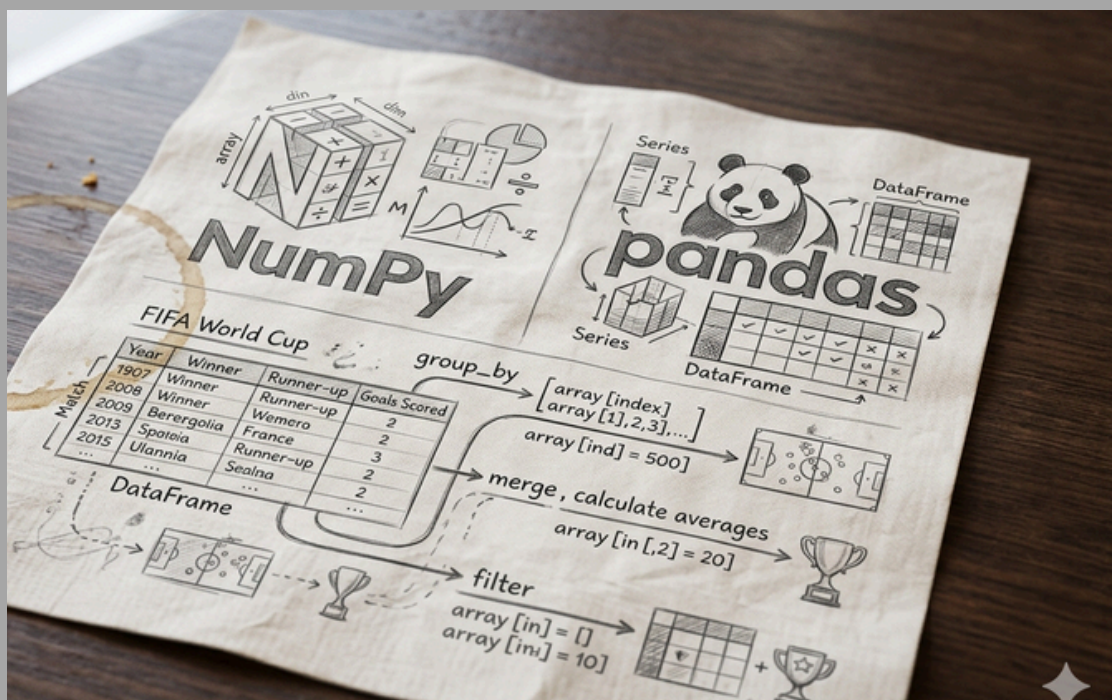
MIT

Academy of Engineering

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FIFA World Cup Data Analysis using NumPy & Pandas



Objective :

To analyze FIFA World Cup dataset using NumPy and Pandas by solving 20 problem statements.

Problem 1: Display first 10 rows

Output Screenshot:

```
Using Colab cache for faster access to the 'fifa-world-cup' dataset.
Dataset Path: /kaggle/input/fifa-world-cup

Year Datetime Stage Stadium City
1930.0 13 Jul 1930 - 15:00 Group 1 Pocitos Montevideo
1930.0 13 Jul 1930 - 15:00 Group 4 Parque Central Montevideo
1930.0 14 Jul 1930 - 12:45 Group 2 Parque Central Montevideo
1930.0 14 Jul 1930 - 14:50 Group 3 Pocitos Montevideo
1930.0 15 Jul 1930 - 16:00 Group 1 Parque Central Montevideo
1930.0 16 Jul 1930 - 14:45 Group 1 Parque Central Montevideo
1930.0 17 Jul 1930 - 12:45 Group 2 Parque Central Montevideo
1930.0 17 Jul 1930 - 14:45 Group 4 Parque Central Montevideo
1930.0 18 Jul 1930 - 14:30 Group 3 Estadio Centenario Montevideo
1930.0 19 Jul 1930 - 12:50 Group 1 Estadio Centenario Montevideo

Home Team Name Home Team Goals Away Team Goals Away Team Name \
France 4.0 1.0 Mexico
USA 3.0 0.0 Belgium
Yugoslavia 2.0 1.0 Brazil
Romania 3.0 1.0 Peru
Argentina 1.0 0.0 France
Chile 3.0 0.0 Mexico
Yugoslavia 4.0 0.0 Bolivia
USA 3.0 0.0 Paraguay
Uruguay 1.0 0.0 Peru
Chile 1.0 0.0 France
```

```
0 France 4.0 1.0 Mexico
1 USA 3.0 0.0 Belgium
2 Yugoslavia 2.0 1.0 Brazil
3 Romania 3.0 1.0 Peru
4 Argentina 1.0 0.0 France
5 Chile 3.0 0.0 Mexico
6 Yugoslavia 4.0 0.0 Bolivia
7 USA 3.0 0.0 Paraguay
8 Uruguay 1.0 0.0 Peru
9 Chile 1.0 0.0 France

Win conditions Attendance Half-time Home Goals Half-time Away Goals \
0 4444.0 3.0 0.0
1 18346.0 2.0 0.0
2 24059.0 2.0 0.0
3 2549.0 1.0 0.0
4 23409.0 0.0 0.0
5 9249.0 1.0 0.0
6 18306.0 0.0 0.0
7 18306.0 2.0 0.0
8 57735.0 0.0 0.0
9 2000.0 0.0 0.0

Referee Assistant 1 \
0 LOMBARDI Domingo (URU) CRISTOPHE Henry (BEL)
1 MACIAS Jose (ARG) MATEUCCI Francisco (URU)
2 TEJADA Anibal (URU) VALLARINO Ricardo (URU)
3 WARRKEN Alberto (CHI) LANGENUS Jean (BEL)
4 REGO Gilberto (BRA) SAUCEDO Ulises (BOL)
5 CRISTOPHE Henry (BEL) APHESTEGUY Martin (URU)
6 MATEUCCI Francisco (URU) LOMBARDI Domingo (URU)
7 MACIAS Jose (ARG) APHESTEGUY Martin (URU)
8 LANGENUS Jean (BEL) BALWAY Thomas (FRA)
9 TEJADA Anibal (URU) LOMBARDI Domingo (URU)
```

Code:

```
import pandas as pd
import numpy as np
import kagglehub

# Load dataset
path = kagglehub.dataset_download("abecklas/fifa-world-cup")
print("Dataset Path:", path)

# Load CSV files (adjust file name if needed)
matches = pd.read_csv(path + "/WorldCupMatches.csv")

# 1. Display first 10 rows
print(matches.head(10))
```

Code:

```
# 2. Total matches
print("Total Matches:", len(matches))
```

Output Screenshot:

```
.. Total Matches: 4572
```

Problem 3: Total goals

Code:

Output Screenshot:

```
# 3. Total goals
matches['TotalGoals'] = matches['Home Team Goals'] + matches['Away Team Goals']
print("Total Goals:", matches['TotalGoals'].sum())
```

```
Total Goals: 2414.0
```

Problem 4: Average goals per match

code:

Output Screenshot:

```
# 4. Average goals per match
print("Average Goals:", matches['TotalGoals'].mean())
```

```
Average Goals: 2.8333333333333335
```

Problem 5: Team with most wins

Code:

Output Screenshot:

```
# 5. Team with most wins
wins = matches['Home Team Name'].value_counts() + matches['Away Team Name'].value_counts()
print("Top Team:", wins.idxmax())
```

```
Top Team: Brazil
```

Problem 6: Year with highest goals

Code:

Output Screenshot:

```
# 6. Year with highest goals
goals_per_year = matches.groupby('Year')['TotalGoals'].sum()
print("Year with max goals:", goals_per_year.idxmax())
```

```
Year with max goals: 2014.0
```

Problem 7: Matches per year

Code:

Output Screenshot:

```
# 7. Matches per year
print(matches.groupby('Year').size())
```

```
Year
1930.0    18
1934.0    17
1938.0    18
1950.0    22
1954.0    26
1958.0    35
1962.0    32
1966.0    32
1970.0    32
1974.0    38
1978.0    38
1982.0    52
1986.0    52
1990.0    52
1994.0    52
1998.0    64
2002.0    64
2006.0    64
2010.0    64
2014.0    80
dtype: int64
```

Problem 8: Top 5 teams by goals

Code:

Output Screenshot:


```
# 8. Top 5 teams by goals
```

```
team_goals = matches.groupby('Home Team Name')['Home Team Goals'].sum()  
print(team_goals.sort_values(ascending=False).head())
```

```
Home Team Name  
Brazil          180.0  
Argentina       111.0  
Italy           99.0  
Germany FR      99.0  
Hungary         73.0  
Name: Home Team Goals, dtype: float64
```

Problem 9: Highest attendance match

Code:

Output Screenshot:

```
# 9. Highest attendance match
```

```
print(matches.loc[matches['Attendance'].idxmax()])
```

```
Year          1950.0  
Datetime      16 Jul 1950 - 15:00  
Stage         Group 6  
Stadium       Maracanã - Estádio Jornalista Mário Filho  
City          Rio De Janeiro  
Home Team Name Uruguay  
Home Team Goals      2.0  
Away Team Goals      1.0  
Away Team Name      Brazil  
Win conditions  
Attendance      173850.0  
Half-time Home Goals      0.0  
Half-time Away Goals      0.0  
Referee        READER George (ENG)  
Assistant 1     ELLIS Arthur (ENG)  
Assistant 2     MITCHELL George (SCO)  
RoundID        209.0  
MatchID        1190.0  
Home Team Initials      URU  
Away Team Initials      BRA  
TotalGoals          3.0  
Name: 74, dtype: object
```

10. Find the average attendance per match

Code:

Output Screenshot:

```
print("Avg Attendance:", matches['Attendance'].mean())
```

```
... Avg Attendance: 45164.8
```

Problem 11: Draw matches

Code:

Output Screenshot:

```
# 11. Draw matches
draws = matches[matches['Home Team Goals'] == matches['Away Team Goals']]
print("Draw Matches:", len(draws))
```

```
Draw Matches: 190
```

12. Find the most frequent stadium used.

Code:

Output Screenshot:

```
# 12. Most frequent stadium
print(matches['Stadium'].value_counts().idxmax())
```

```
Estadio Azteca
```

Problem 13: Win percentage

Code:

Output Screenshot:

```
# 13. Win percentage (home team approx)
home_wins = matches[matches['Home Team Goals'] > matches['Away Team Goals']]
win_percent = len(home_wins) / len(matches) * 100
print("Home Win %:", win_percent)
```

```
Home Win %: 10.673665791776028
```

14. Identify the host country that hosted the most tournaments.

Code:

Output Screenshot:

```
# 14. Host country with most tournaments
worldcups = pd.read_csv(path + "/WorldCups.csv")
print(worldcups['Country'].value_counts().idxmax())

Italy
```

Problem 15: Highest scoring match

Code:

Output Screenshot:

```
# 1. Clean the data (the FIFA dataset often has empty rows at the bottom)
matches = matches.dropna(subset=['Home Team Goals', 'Away Team Goals'])

# 2. Create the 'TotalGoals' column by adding the home and away scores
# Make sure the column names match your CSV exactly
matches['TotalGoals'] = matches['Home Team Goals'] + matches['Away Team Goals']

# 3. Find the index of the highest value and locate that row
highest_scoring_match = matches.loc[matches['TotalGoals'].idxmax()]

# 4. Print the result
print("--- Highest Scoring Match Details ---")
print(highest_scoring_match)

--- Highest Scoring Match Details ---
Year                1954.0
Datetime            26 Jun 1954 - 17:00
Stage               Quarter-finals
Stadium             La Pontaise
City                Lausanne
Home Team Name      Austria
Home Team Goals     7.0
Away Team Goals     5.0
Away Team Name      Switzerland
Win conditions
Attendance          35000.0
Half-time Home Goals 5.0
Half-time Away Goals 4.0
Referee             FAULTLESS Charlie (SCO)
Assistant 1         ASENSI Manuel (ESP)
Assistant 2         SCHMETZER Emil (FRG)
RoundID             212.0
MatchID             1237.0
Home Team Initials  AUT
Away Team Initials  SUI
TotalGoals          12.0
Name: 94, dtype: object
```

16. Calculate the standard deviation of goals scored.

Code:

Output Screenshot:

```
# 16. Standard deviation of goals
print("Std Dev Goals:", np.std(matches['TotalGoals']))

Std Dev Goals: 1.953188477059263
```

Problem 17: Matches above average goals

Code:

Output Screenshot:

```
# 17. Matches above average goals
avg_goals = matches['TotalGoals'].mean()
print(matches[matches['TotalGoals'] > avg_goals])
```

	Year		Datetime	Stage \
0	1930.0	13 Jul 1930	- 15:00	Group 1
1	1930.0	13 Jul 1930	- 15:00	Group 4
2	1930.0	14 Jul 1930	- 12:45	Group 2
3	1930.0	14 Jul 1930	- 14:50	Group 3
5	1930.0	16 Jul 1930	- 14:45	Group 1
..	...		...	...
841	2014.0	30 Jun 2014	- 17:00	Round of 16
843	2014.0	01 Jul 2014	- 17:00	Round of 16
845	2014.0	04 Jul 2014	- 17:00	Quarter-finals
848	2014.0	08 Jul 2014	- 17:00	Semi-finals
850	2014.0	12 Jul 2014	- 17:00	Play-off for third place

	Stadium	City	Home Team Name	Home Team Goals \
0	Pocitos	Montevideo	France	4.0
1	Parque Central	Montevideo	USA	3.0
2	Parque Central	Montevideo	Yugoslavia	2.0
3	Pocitos	Montevideo	Romania	3.0
5	Parque Central	Montevideo	Chile	3.0
..	...	...	...	...
841	Estadio Beira-Rio	Porto Alegre	Germany	2.0
843	Arena Fonte Nova	Salvador	Belgium	2.0
845	Estadio Castelao	Fortaleza	Brazil	2.0
848	Estadio Mineirao	Belo Horizonte	Brazil	1.0
850	Estadio Nacional	Brasilia	Brazil	0.0

	Away Team Goals	Away Team Name	Win conditions ... \
0	1.0	Mexico	...
1	0.0	Belgium	...
2	1.0	Brazil	...
3	1.0	Peru	...
5	0.0	Mexico	...

..	...	...	...	...
841	1.0	Algeria	Germany win after extra time	...
843	1.0	USA	Belgium win after extra time	...
845	1.0	Colombia		...
848	7.0	Germany		...
850	3.0	Netherlands		...

	Half-time Home Goals	Half-time Away Goals	\
0	3.0	0.0	
1	2.0	0.0	
2	2.0	0.0	
3	1.0	0.0	
5	1.0	0.0	
..	...	...	
841	0.0	0.0	
843	0.0	0.0	
845	1.0	0.0	
848	0.0	5.0	
850	0.0	2.0	

	Referee	Assistant 1	\
0	LOMBARDI Domingo (URU)	CRISTOPHE Henry (BEL)	
1	MACIAS Jose (ARG)	MATEUCCI Francisco (URU)	
2	TEJADA Anibal (URU)	VALLARINO Ricardo (URU)	
3	WARNKEN Alberto (CHI)	LANGENUS Jean (BEL)	
5	CRISTOPHE Henry (BEL)	APHESTEGUY Martin (URU)	
..	...	...	
841	RICCI Sandro (BRA)	DE CARVALHO Emerson (BRA)	
843	HAIMOUDI Djamel (ALG)	ACHIK Redouane (MAR)	
845	Carlos VELASCO CARBALLO (ESP)	ALONSO FERNANDEZ Roberto (ESP)	
848	RODRIGUEZ Marco (MEX)	TORRETERA Marvin (MEX)	
850	HAIMOUDI Djamel (ALG)	ACHIK Redouane (MAR)	

	Assistant 2	RoundID	MatchID	Home Team Initials	\
0	REGO Gilberto (BRA)	201.0	1096.0	FRA	
1	WARNKEN Alberto (CHI)	201.0	1090.0	USA	
2	BALWAY Thomas (FRA)	201.0	1093.0	YUG	
3	MATEUCCI Francisco (URU)	201.0	1098.0	ROU	
5	LANGENUS Jean (BEL)	201.0	1095.0	GUE	

2	BALWAY Thomas	(FRA)	201.0	1093.0	YUG
3	MATEUCCI Francisco	(URU)	201.0	1098.0	ROU
5	LANGENUS Jean	(BEL)	201.0	1095.0	CHI
..	...	...	...	...	...
841	VAN GASSE Marcelo	(BRA)	255951.0	300186460.0	GER
843	ETCHIALI Abdelhak	(ALG)	255951.0	300186497.0	BEL
845	YUSTE Juan	(ESP)	255953.0	300186461.0	BRA
848	QUINTERO Marcos	(MEX)	255955.0	300186474.0	BRA
850	ETCHIALI Abdelhak	(ALG)	255957.0	300186502.0	BRA
Away Team Initials TotalGoals					
0	MEX	5.0			
1	BEL	3.0			
2	BRA	3.0			
3	PER	4.0			
5	MEX	3.0			
..	...	...			
841	ALG	3.0			
843	USA	3.0			
845	COL	3.0			
848	GER	8.0			
850	NED	3.0			
[446 rows x 21 columns]					

18.Determine the most common match outcome (home win, away win, draw)

Code:

Output Screenshot:

```

import pandas as pd
import numpy as np

# 1. Ensure we drop empty rows to avoid logic errors with NaN
matches = matches.dropna(subset=['Home Team Goals', 'Away Team Goals'])

# 2. Define your conditions
conditions = [
    (matches['Home Team Goals'] > matches['Away Team Goals']),
    (matches['Home Team Goals'] < matches['Away Team Goals']),
    (matches['Home Team Goals'] == matches['Away Team Goals'])
]

# 3. Define your choices
choices = ['Home Win', 'Away Win', 'Draw']

# 4. Use np.select with a default value to prevent dtype conflicts
# We use default='Unknown' to ensure everything is treated as a string
matches['Result'] = np.select(conditions, choices, default='Unknown')

# 5. Print the counts
print(matches['Result'].value_counts())

```

```

Result
Home Win    488
Draw        190
Away Win    174
Name: count, dtype: int64

```

Problem 19: Correlation analysis

Code:

Output Screenshot:

```

# 19. Correlation between attendance and goals
print(matches[['Attendance', 'TotalGoals']].corr())

```

	Attendance	TotalGoals
Attendance	1.000000	-0.113603
TotalGoals	-0.113603	1.000000

20. Use NumPy to normalize attendance values.

Code:

Output Screenshot:

```
# 20. Normalize attendance (Min-Max scaling)
attendance = matches['Attendance']
normalized = (attendance - np.min(attendance)) / (np.max(attendance) - np.min(attendance))

matches['Normalized Attendance'] = normalized
print(matches[['Attendance', 'Normalized Attendance']].head())
```

	Attendance	Normalized Attendance
0	4444.0	0.014222
1	18346.0	0.095118
2	24059.0	0.128362
3	2549.0	0.003195
4	23409.0	0.124580